PWM programming.

Pulse Width Modulation Basics

Pulse Width Modulation or PWM is a way to encode data such that it corresponds to the width of the pulse given a fixed frequency. It is also a way to control motors, power circuits, etc., using the ‘width’ of the pulse. PWM has numerous uses like Motion Control, Dimming, Encoding Analog Signal into its Digital form, in Power Regulation , etc.

The PWM is based on the standard Timer block and inherits all of its features, although only the PWM function is pinned out on the LPC1768.

The Timer is designed to count cycles of the peripheral clock (PCLK) and optionally generate interrupts or perform other actions when specified timer values occur, based on seven match registers. The PWM function is in addition to these features and is based on match register events.

PWM can be used for more applications. For instance, multi-phase motor control typically requires three non-overlapping PWM outputs with individual control of all three pulse widths and positions.

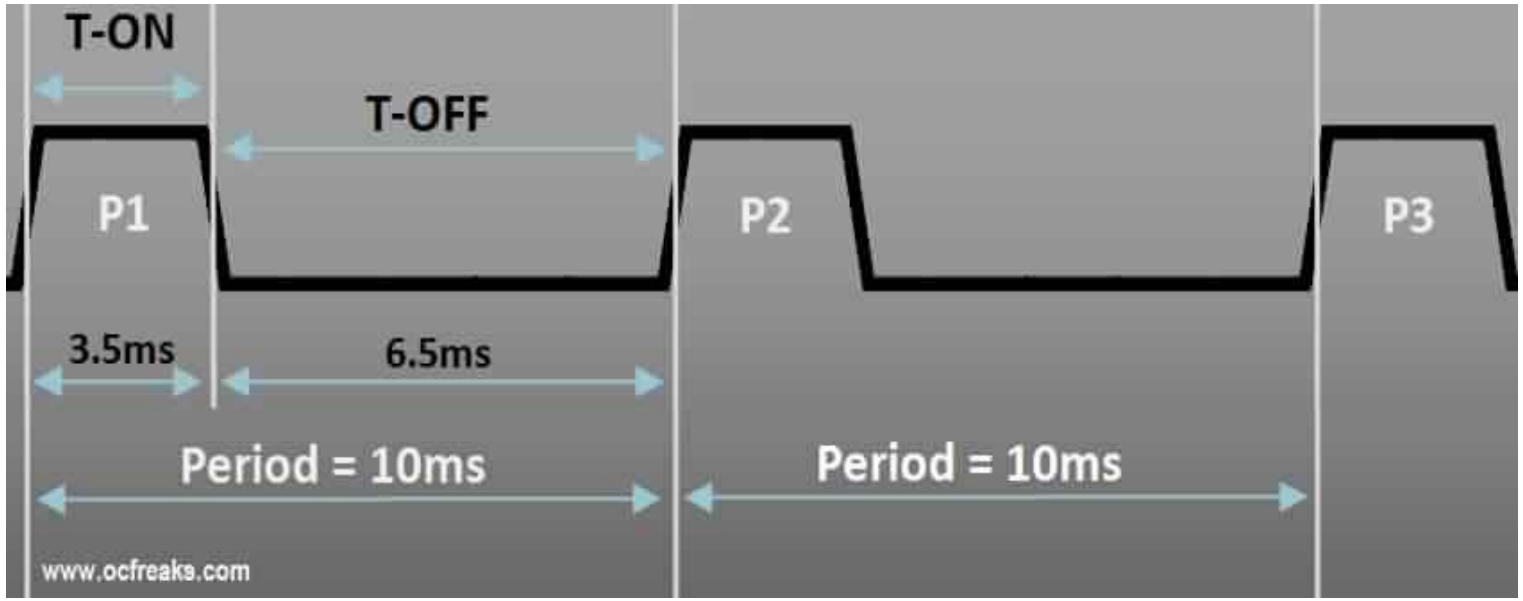
PWM Edges

A PWM signal contains 2 types of Edges which are called Leading Edge and Trailing Edge. The Diagram shown below explains this:

T-ON , T-OFF & DutyCycle :

$$\text{Duty Cycle} = \frac{\text{T-ON}}{\text{T-ON + T-OFF}}$$

$$\text{Duty Cycle \%} = \frac{\text{T-ON}}{\text{T-ON + T-OFF}} \text{X100}$$



Period i.e T = 10ms
Frequency = 100hz
T-ON = 3.5ms
T-OFF = 6.5ms
DutyCyle = 35%

PWM Edges

A PWM signal contains 2 types of Edges which are called Leading Edge and Trailing Edge. The Diagram shown below explains this:

Types of PWM

PWM Signal can be Classified in Different ways.

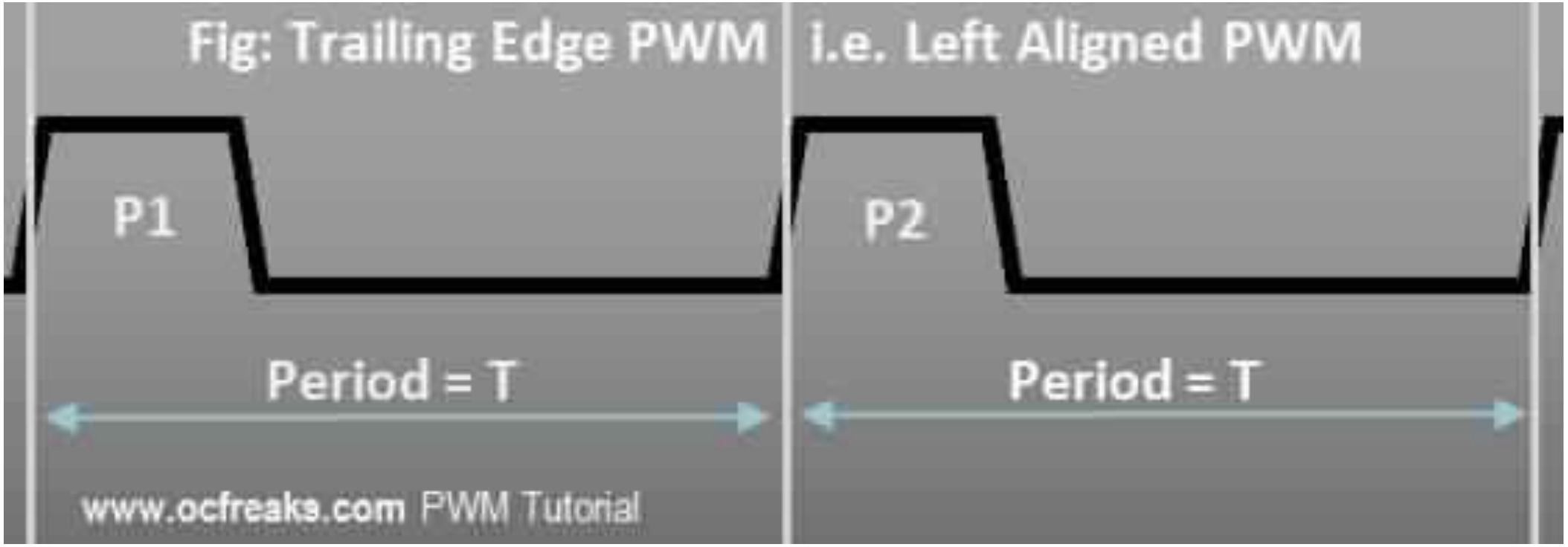
1) **Single Edge PWM**

2) **Double Edge PWM**

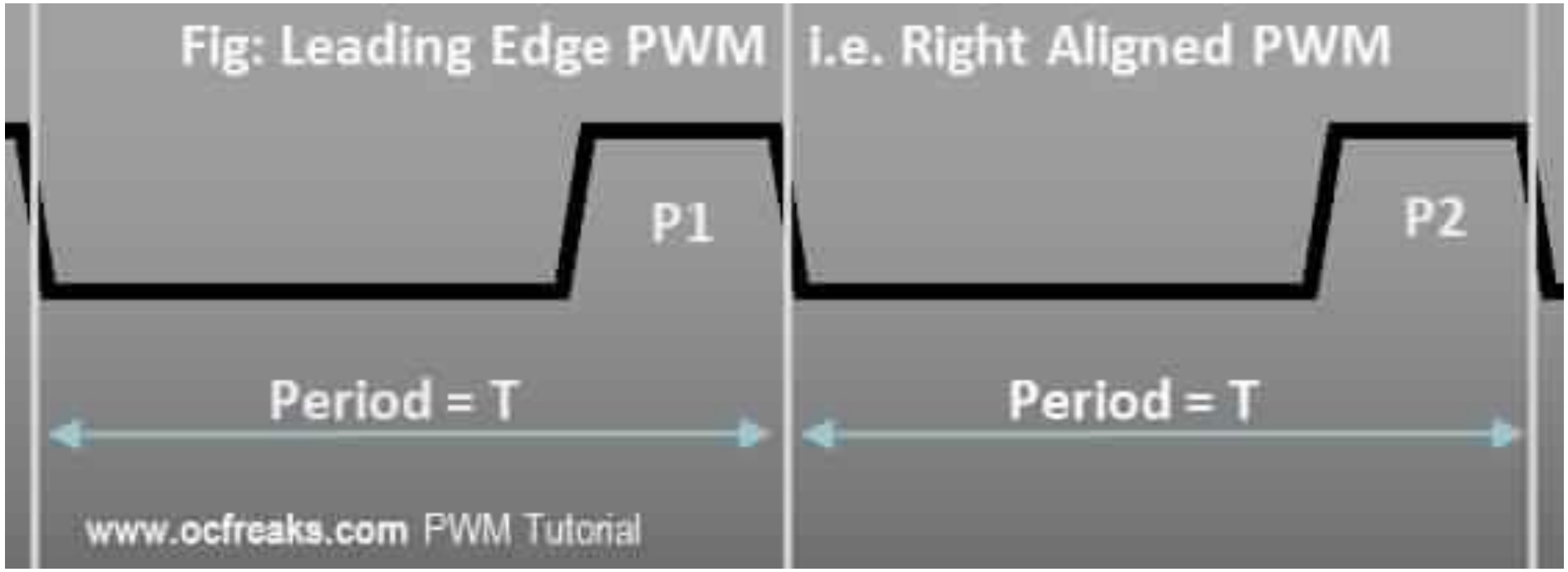
Single Edge PWM

In Single Edge PWM, the Pulse can either at the Beginning or the End of the Period and hence can be further Classified into **Leading Edge(Right Aligned) PWM and Trailing Edge(Left Aligned) PWM**.

In Trailing Edge PWM the Leading Edge is fixed at the Beginning of a Period and the **Trailing Edge is Modulated i.e. Varied**.

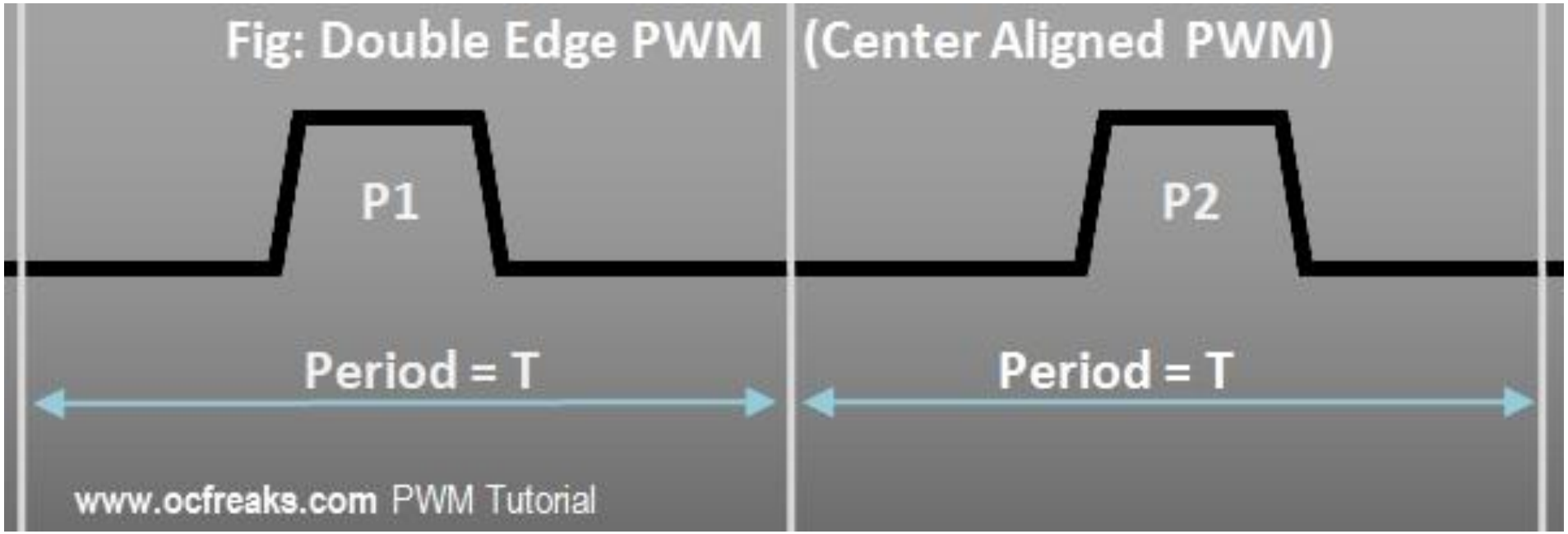


In Leading Edge PWM the Trailing Edge is fixed at the End of a Period and the **Leading Edge is Modulated i.e. Varied**.



Double Edge PWM

In Double Edge PWM the Pulse can be Positioned anywhere within the Period. Its called “Double Edge” because both the Edges are Modulated or Varied. In applications like Multi-Phase Motor control Double Edge PWM is used where the Pulse is Center Aligned to reduce Harmonics.



PWM Voltage

The **Average Voltage** of a PWM Output depends on its Duty Cycle. Lower the Duty Cycle lower will be the Voltage and Vice-Versa.

$$\mathbf{V_{average} = DutyCycle \times V_H}$$

In case the Low State Represents a Negative Voltage then the above equation can be generalized as follows:

$$\mathbf{V_{average} = (DutyCycle \times V_H) + ((1-D) \times V_L)}$$

Where , V_H = Voltage for High State & V_L = Voltage for Low State

ARM Cortex-M3 LPC1768 PWM Module

The LPC1768 Contains a single PWM Module called PWM1 and supports 2 types of PWM:

1) Single Edge PWM – Pulse starts with new Period i.e., Pulse is always at the beginning

2) Double Edge PWM – Pulse can be present anywhere within the Period

A PWM block, similar to a Timer block, has a Timer Counter and an associated Prescale Register along with Match Registers. These work exactly the same way as in the case of Timers.

Match Registers 1 to 6 (except 0) are pinned on LPC1768 i.e., the corresponding outputs are given to actual Pins on LPC1768 MCU. The PWM function must be selected for these Pins, using PINSELx Register, to get the PWM output. These pins are:

Output	PWM1.1	PWM1.2	PWM1.3	PWM1.4	PWM1.5	PWM1.6
Pin Name	P1.18 P2.0	P1.20 P2.1 P3.25	P1.21 P2.2 P3.26	P1.23 P2.3	P1.24 P2.4	P1.26 P2.5

PWM1.1 output corresponds to PWM Match Register 1 i.e., PWM1MR1 , PWM1.2 output corresponds to PWM1MR2 , and so on. Also Match Register 0 i.e., PWM1MR0 is NOT pinned because it is used to generate the PWM Period. In LPC1768, PWM1.3 is not pinned out.

There are 7 match registers inside the PWM1 block. The first Match register PWM1MR0 is used to generate PWM period and hence we are left with 6 Match Registers PWM1MR1 to PWM1MR6 to generate 6 Single Edge PWM signals or 3 Double Edge PWM signals. Double edge PWM uses 2 match registers hence we can get only 3 double edge outputs.

PWM Rules –

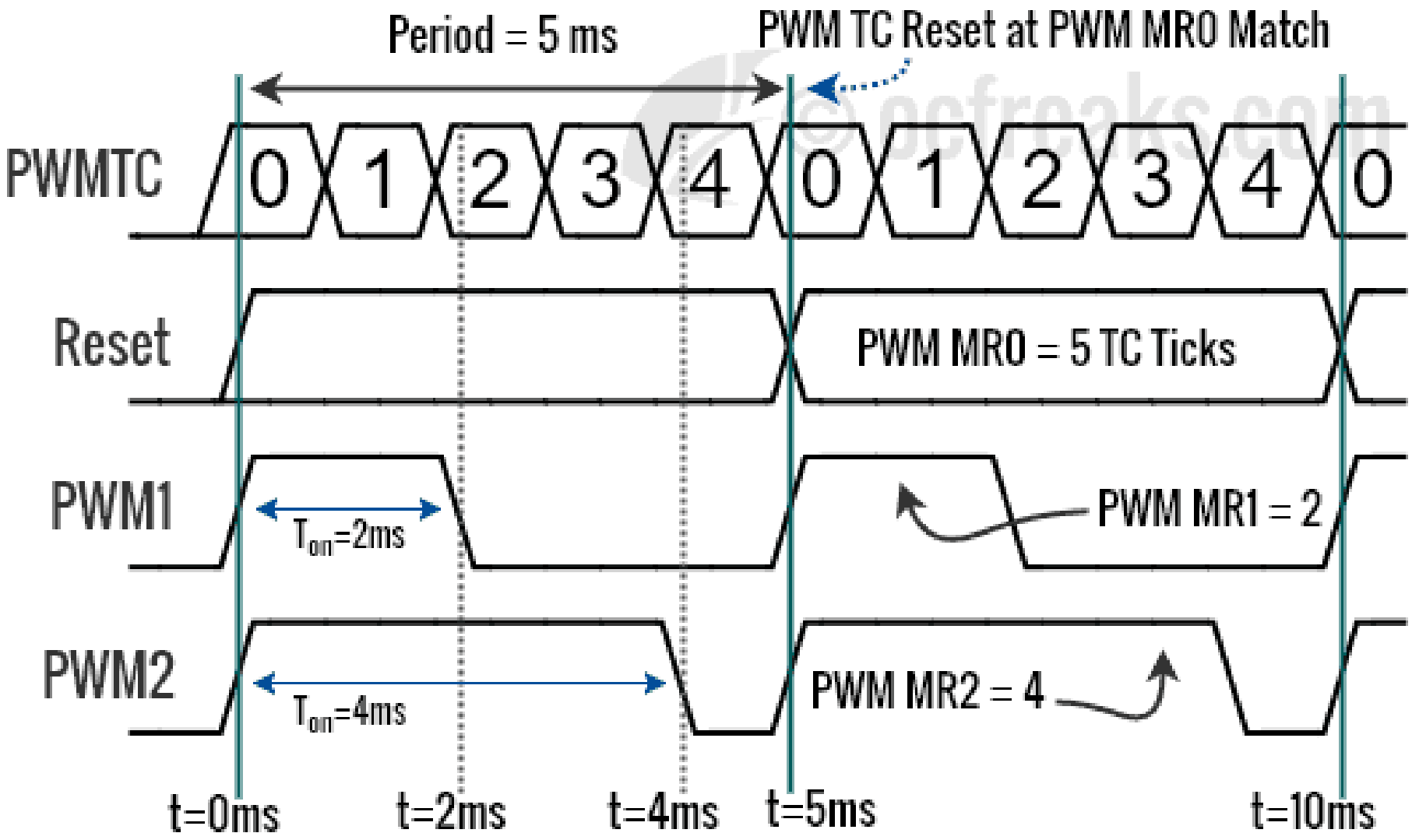
•The Rules for Single Edged PWM are:

1) All single edged PWM outputs will go high at the beginning of a PWM cycle unless their match value is 0.

2) Each PWM output will go low when its match value is reached.

Note: If no match occurs i.e Match value is greater than Period then the output will remain high!

Consider that our PWM Period duration is 5 milliseconds and TC increments every 1 millisecond using appropriate prescale value. We set PWM1MR0 to 5 i.e. **5 ticks of PWM TC**. We have PWM1MR1 = 2 and PWM1MR2 = 4. Whenever the value in TC matches the value in any of the match register, its corresponding output is set to low until the start of next PWM cycle as shown in PWM Timing Diagram below:


Period = 5 ms
PWM TC Reset at PWM MR0 Match
PWMTC
0
1
2
3
4
0
1
2
3
4
0
Reset
PWM MR0 = 5 TC Ticks
PWM1
T_{on}=2ms
PWM MR1 = 2
PWM2
T_{on}=4ms
PWM MR2 = 4
t=0ms
t=2ms
t=4ms
t=5ms
t=10ms

Registers used in LPC176x PWM Programming

1. PWM1TCR – PWM Timer Control Register: This register is used to control the Timer Counter inside the PWM block. Only Bits: 0, 1 & 3 are used rest are reserved.

•**Bit 0 :** When 1 both PWM Timer counter and PWM Prescale counter are enabled. When 0 both are disabled.

•**Bit 1 :** When set to 1 it will reset both Timer and Prescale counter of PWM block (at next edge of PCLK).

Bit 3 : Enables PWM mode i.e., the PWM outputs. Note that the PWM Match register that determines the PWM rate (PWM Match Register 0 - MR0) must be set up prior to the PWM being enabled.

•Other Bits : Reserved.

2) PWM1PR – PWM Prescale Register: PWMPR is used to control the resolution of the PWM outputs. The Timer Counter(TC) will increment every PWMPR+1 Peripheral Clock Cycles (PCLK).

3) PWM1MR0 – PWM1MR6 (Match Registers): The seven Match registers contain **Pulse Width Values i.e the Number of PWM1TC Ticks**. **PWM1MR0** holds the value of 1 complete cycle or the period.

4) PWM1MCR – PWM Match Control Registers: The PWM Match Control Register is used to specify what operations can be done when the value in a particular Match register equals the value in TC. For each Match Register we have 3 options : Either generate an Interrupt , or Reset the TC, or Stop, which stops the counters and disables PWM. Hence this register is divided into group of 3 bits. The first 3 bits are for Match Register 0 i.e PWMMR0 , next 3 for PWMMR1 , and so on.

1) Bit 0 : Interrupt on PWM1MR0 Match – If set to 1 then it will generate an Interrupt else disable if set to 0.

2) Bit 1 : Reset on PWM1MR0 Match – If set to 1 it will reset the PWM Timer Counter i.e. PWM1TC else disabled if set to 0.

3) Bit 2 : Stop on PWM1MR0 Match – If this bit is set 1 then both PWM1TC and PWM1PC will be stopped & disable the Counters.

4) Similarly {Bits 3,4,5} for PWM1MR1 , {Bits 6,7,8} for PWM1MR2 , {Bits 9,10,11} for PWM1MR3 ,{Bits 12,13,14} for PWM1MR4 ,{Bits 15,16,17} for PWM1MR5 , {Bits 18,19,20} for PWM1MR6.

5) PWM1IR – PWM Interrupt Register: If an interrupt is generated by any of the Match Register then the corresponding bit in PWM1IR will be set high. Otherwise, the bit will be low.

1) Bits 0,1,2,3 are for PWM1MR0, PWM1MR1, PWM1MR2, PWM1MR3 respectively and

2) Bits 8,9,10 are for PWM1MR4 , PWM1MR5 , PWM1MR6 respectively. Other bits are reserved.

6) PWMLER – Load Enable Register: The PWM Load Enable Register is used to control the way Match Registers are updated when PWM generation is active. A program write to a Match register will not have an effect on the Match result until the corresponding bit in PWMLER has been set, followed by the occurrence of a PWM Match 0 event.
Each Match Register has a corresponding Shadow Register. When a PWM Match 0 event occurs (normally also resetting the timer in PWM mode), the contents of match registers will be transferred to the shadow registers if the corresponding bit in the Latch Enable Register has been set. At that point, the new values will take effect and determine the course of the next PWM cycle. Once the transfer of new values has taken place, all bits of the LER are automatically cleared. Until the corresponding bit in the PWMLER is set and a PWM Match 0 event occurs, any value written to the PWM Match registers has no effect on PWM operation.

Bit ‘x’ in PWM1LER corresponds to match Register ‘x’. i.e., Bit 0 is for PWM1MR0 , Bit 1 for PWM1MR1 , .. and so on.

7) PWM1PCR – PWM Control Register: This register is used for Selecting between Single Edged & Double Edged outputs and also to Enable/Disable the 6 PWM outputs which go to their corresponding Pins.

1) **Bits 2 to 6** are used to select between Single or Double Edge mode for PWM 2,3,4,5,6 outputs. If **Bit 2** is set to 1 then PWM1.2 (i.e., the one corresponding to PWM1MR2) output is double edged else if set 0 then its Single Edged. Similarly **Bit x** for PWM1.x correspondingly.

2) **Bits 9 to 14** are used to Enable/Disable PWM outputs. If **Bit 9** is set to 1 then PWM1.1 output is enabled , else disabled if set to 0. Similarly remaining bits for PWM1.x correspondingly.

How to generate PWM?

When the value for the cycle time or the period is loaded into MR0, The selected PWM output goes high. TC starts counting. When it matches with the value in the corresponding match register, the PWM output goes low and remains low until TC matches with the MR0. Then the next cycle starts. For the next cycle if the match register value corresponding to the PWM output is changed, the pulse width is changed.

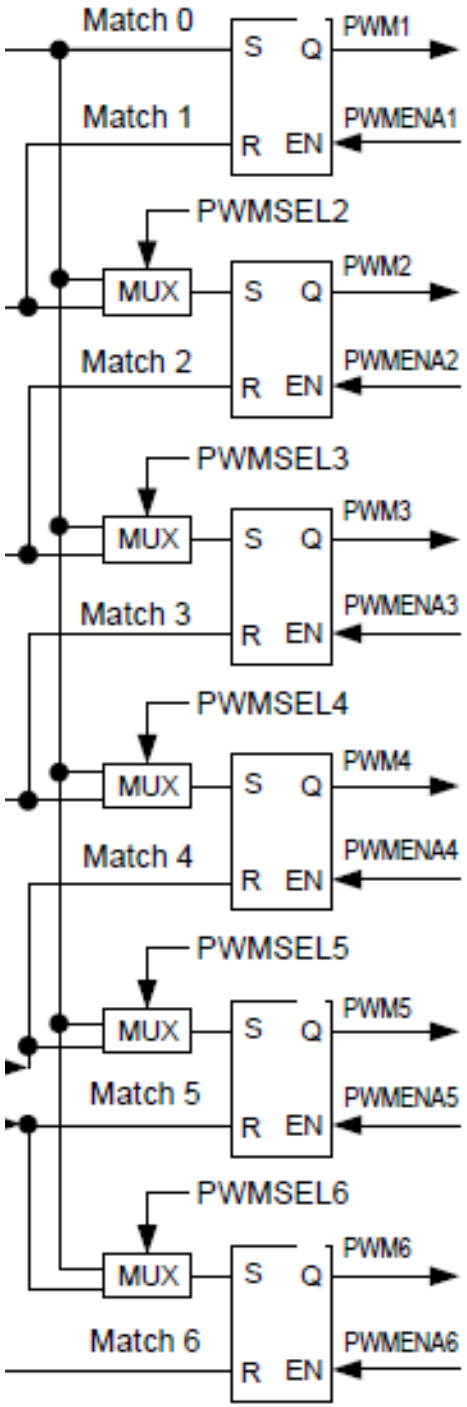
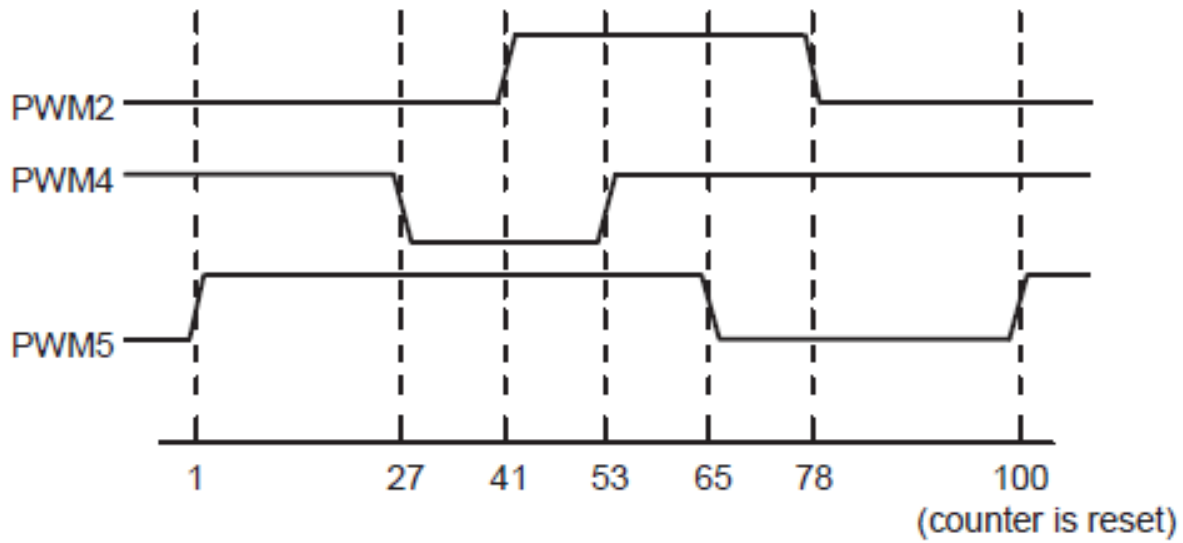


Table 443. Set and reset inputs for PWM Flip-Flops

PWM Channel	Single Edge PWM (PWMSELn = 0)		Double Edge PWM (PWMSELn = 1)	
	Set by	Reset by	Set by	Reset by
1	Match 0	Match 1	Match 0[1]	Match 1[1]
2	Match 0	Match 2	Match 1	Match 2
3	Match 0	Match 3	Match 2[2]	Match 3[2]
4	Match 0	Match 4	Match 3	Match 4
5	Match 0	Match 5	Match 4[2]	Match 5[2]
6	Match 0	Match 6	Match 5	Match 6

[1] Identical to single edge mode in this case since Match 0 is the neighboring match register. Essentially, PWM1 cannot be a double edged output.

[2] It is generally not advantageous to use PWM channels 3 and 5 for double edge PWM outputs because it would reduce the number of double edge PWM outputs that are possible. Using PWM 2, PWM4, and PWM6 for double edge PWM outputs provides the most pairings.



The waveforms show one PWM cycle and demonstrate PWM outputs under the following conditions:

The timer is configured for PWM mode (counter resets to 1).

Match 0 is configured to reset the timer/counter when a match event occurs.

Control bits PWMSEL2 and PWMSEL4 are set.

The Match register values are as follows:

MR0 = 100 (PWM rate)

MR1 = 41, MR2 = 78 (PWM2 output)

MR3 = 53, MR4 = 27 (PWM4 output)

MR5 = 65 (PWM5 output)

Fig 120. Sample PWM waveforms

Configuring and Initializing PWM Block

1. Select the PWM function for the PIN on which you need the PWM output using applicable `LPC_PINCON->PINSELx` register.
2. Select Single Edge or Double Edge Mode using `LPC_PWM1->PCR`. By default its Single Edge Mode.
3. Load the value to `LPC_PWM1->PR`.
4. Set the Value for PWM Period in `LPC_PWM1->MR0`.
5. Set the Values for other Match Registers i.e the Pulse Widths.
6. Set appropriate bit values in `LPC_PWM1->MCR`, like for e.g. resetting PWM1TC for PWMMR0 match and optionally generate interrupts if required
7. Set Load Enable Bits for the Match Registers that you've used. This is important!
8. Then Enable PWM outputs using `LPC_PWM1->PCR`.
9. Now Reset PWM Timer using `LPC_PWM1->TCR`.
10. Finally, Enable Timer Counter and PWM Mode using `LPC_PWM1->TCR`.

Ref:

http://www.ocfreaks.com/pulse-width-modulation-pwm-tutorial/

http://www.ocfreaks.com/lpc1768-pwm-programming-tutorial/

https://www.exploreembedded.com/wiki/LPC1768:_PWM